

TECHNICAL DOCUMENT

LTspice Simulation Methodology

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LTspice Simulation Methodology

Common setup

- Simulator: LTspice 26.0.2 for Windows.
- Device model: PTM 65-nm bulk CMOS beta model released 22 February 2006.
- Nominal condition: 1.2 V, 27 C, 125 MHz, 10 fF per observed output.
- NMOS: L=65 nm, W=120 nm.
- PMOS: L=65 nm, W=240 nm.
- Input edge time: 20 ps.
- XOR transient maximum timestep: 2 ps.
- Encoder transient maximum timestep: 20 ps.
- Power: average of $-V(VDD)*I(VSUP)$ over the documented steady-state window.
- Delay: 50%-to-50%; $tpd=(tpLH+tpHL)/2$.
- Extended encoder delay endpoint: overall parity P0.

Sweeps

The experiment set uses independent one-factor sweeps around the nominal condition: VDD = 0.6, 0.8, 1.0, 1.2 V; temperature = -20, 27, 85 C; load = 1, 5, 10, 20 fF; frequency = 25, 125, 200 MHz. A redundant full Cartesian product is intentionally avoided.

Functional criteria

For the automated operating map, a measured point is considered functionally acceptable when VOH is at least 0.8 VDD, VOL is at most 0.2 VDD, and propagation delay is less than half the input period. This is an explicit study criterion, not a claim that it is a foundry-qualified noise-margin specification.

Traceability

Every generated netlist and log has an experiment ID. The manifest is ``data/experiments.json``; raw LTspice logs stay next to their circuits; parsed results are in ``data/processed/results.csv``; web data are in ``thesis/data/results.json``.

Measurement Definitions

- **Average power:** time average of positive power drawn from VSUP, calculated as $AVG(-V(VDD)*I(VSUP))$.
- **tpLH / tpHL:** elapsed time from the 50% input crossing to the corresponding 50% output crossing.
- **Propagation delay:** arithmetic mean of tpLH and tpHL.
- **PDP:** average power multiplied by propagation delay; recalculated in Python rather than copied manually.
- **Energy per encoding operation:** average encoder power divided by the input-word rate used by the activity pattern.

- **Energy per encoded bit:** energy per operation divided by 11 or 12, consistently by codeword length.
- **Output swing:** maximum measured output minus minimum measured output.
- **Hardware complexity:** transistor count from the instantiated cell topology. It is not layout area.

Official Software Source

LTspice 26.0.2 was obtained from <https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>. The project uses .param, .temp, .tran, and .meas directives in retained circuit files.